MODULE 2 · STATISTICS & PROBABILITY · TOPIC 2.7

The Central Limit Theorem

Day 27 — the result everything else in this module rests on



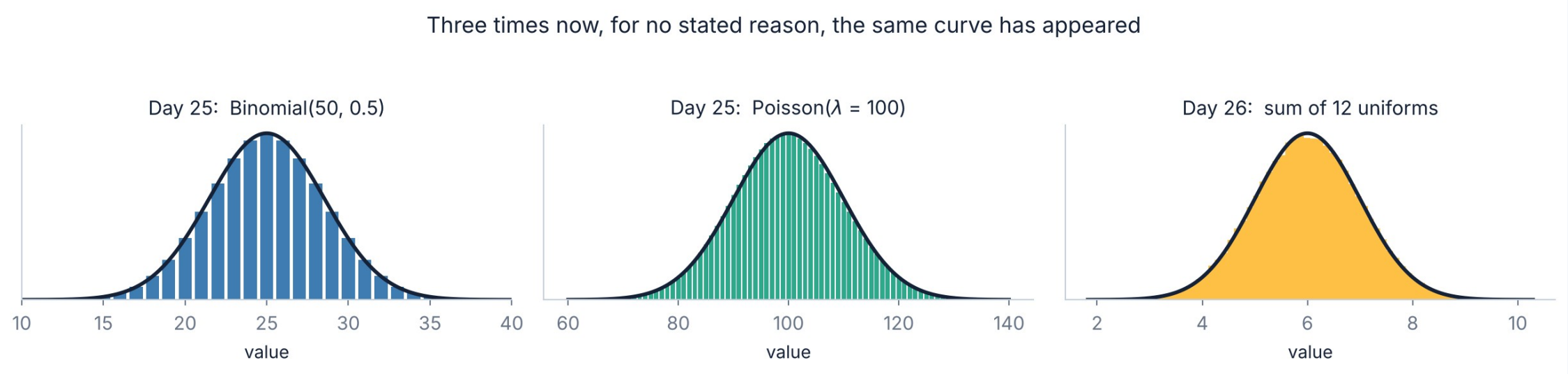
Boston University

3	1	2
Sessions have been setting this up	Distinction that decides everything	Ways it still fails

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THE SETUP

You have already seen this happen three times



Three unrelated distributions, one curve

A binomial counts successes out of n. A Poisson counts events in an interval. A uniform is a flat line with nothing interesting about it at all. They share no mechanism, no shape and no parameters.

Each one turned into the same bell curve, and each time the session ended by saying “Day 27 explains why”. This is Day 27.

WARM-UP · 5 MINUTES

Four questions from Day 26

1	**Why is P(X = exactly 5) zero?**	→ Probability is area, and a point has no width.
2	**U(a, b) in scipy — write it.**	→ uniform(a, b − a). The second argument is a width.
3	**sd of U(0,1)?**	→ 1/√12 = 0.2887.
4	**Twelve uniforms added together gave...**	→ A bell curve. Nobody said why.

Question 4 is today's whole session. And the answer is bigger than it looks: nearly every confidence interval, p-value and A/B test you will ever run is standing on it.

SESSION PLAN

How today's two hours are structured

1. **The one distinction** — 20 min
   Your data is not your sample mean

2. **The theorem** — 25 min
   What it says, and the √n law

3. **How large is large enough?** — 20 min
   The n = 30 rule is not true

4. **When it fails** — 15 min
   Infinite variance, and dependence

5. **What it buys you** — 20 min
   A worked confidence interval

6. **Practice notebook** — 30 min
   You code, I walk around

Section 1 is short and it is the one that matters. Almost every mistake people make with this theorem comes from confusing two things that the next slide separates.

1

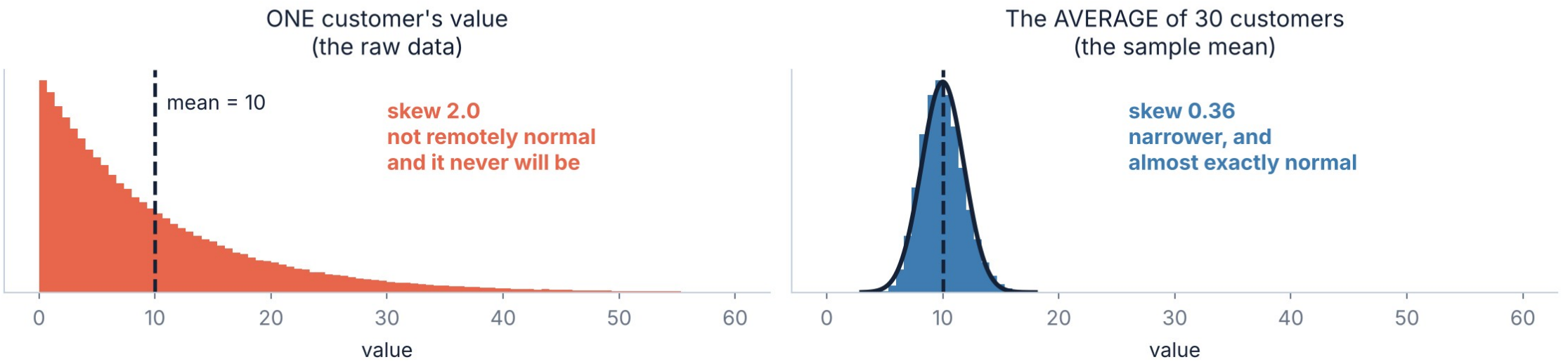
The One Distinction

Your data is not your sample mean · 20 minutes

- Two different distributions
- What the theorem does NOT say

⭐ Two different distributions, and people confuse them

Same population, same axis. These are two completely different distributions.



Left: the distribution of the DATA

What a single observation looks like. Collect a million of them and this shape does not change.

Right: the distribution of the MEAN

What you would get if you took the average of 30, over and over. This is what the theorem is about.

⚠️ What the theorem does NOT say

✕ The three wrong versions

“Big samples are normally distributed”

No. A million salaries are just as skewed as a hundred.

“With n > 30 I can assume normality”

Of the MEAN, sometimes. Never of the data.

“Everything in nature is normal”

Income, city size and file size are all famously not.

✓ What it actually says

Take a sample of n, compute its mean, and write that one number down. Do it again. And again.

The collection of those MEANS is normal — almost regardless of what the data looked like.

The data stays exactly as skewed as it always was. Nothing happens to it at all.

How to catch yourself making this mistake

If you ever find yourself running a normality test on your raw data and then citing the CLT when it fails — stop. The CLT was never a claim about your raw data. It would not have made that data normal, and it does not need to.

2

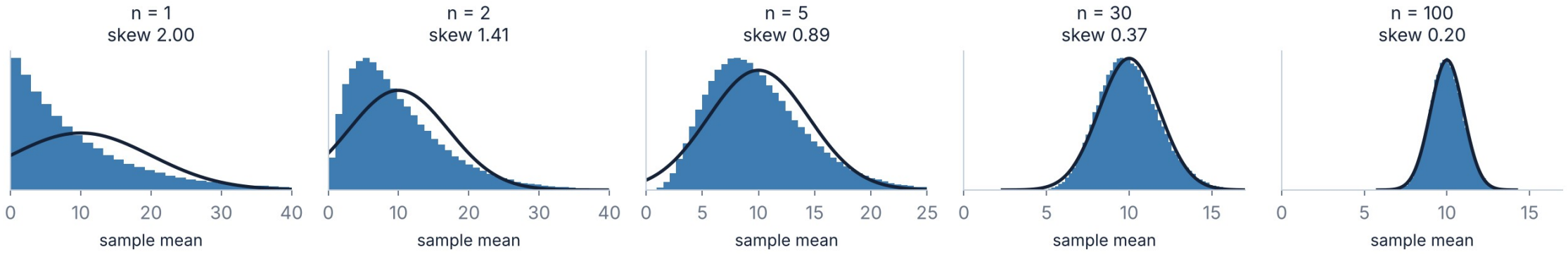
The Theorem

What it says, and what it costs · 25 minutes

- Watch it happen
- The statement
- Standard error
- The √n law

Watch it happen, one sample size at a time

The population is a hard-skewed exponential throughout. Only the sample size changes.



Nothing changes except n

Every panel draws from the same hard-skewed exponential. At n = 1 you are just looking at the raw data again. By n = 30 the black normal curve lies on top of the bars.

And notice the width shrinking at the same time.

Two things are happening at once

The shape becomes normal, AND the spread narrows. They are separate claims and the theorem makes both.

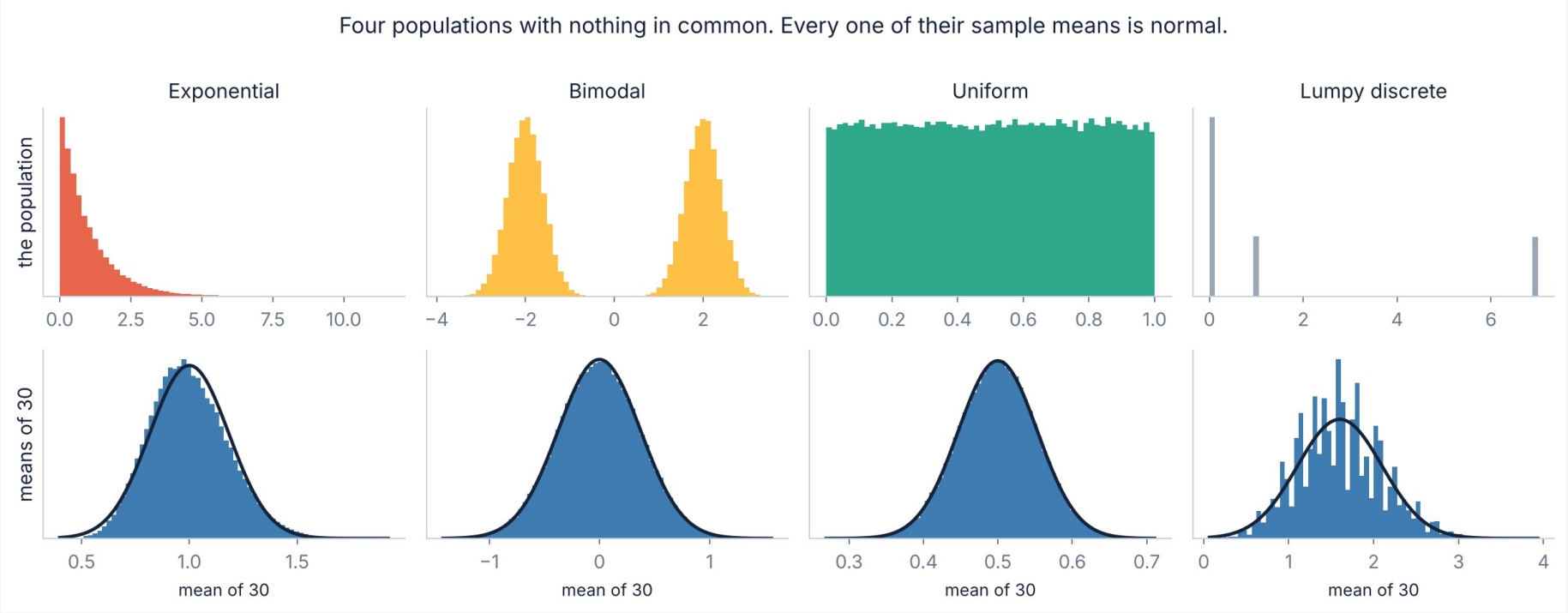
The skew number under each panel is not measured — it is predicted exactly, and section 3 gives the formula.

The theorem, in one sentence

Take samples of size n from any population with a finite mean μ and a finite standard deviation σ. Then the sample means are normal, centred on μ, with standard deviation σ/√n.

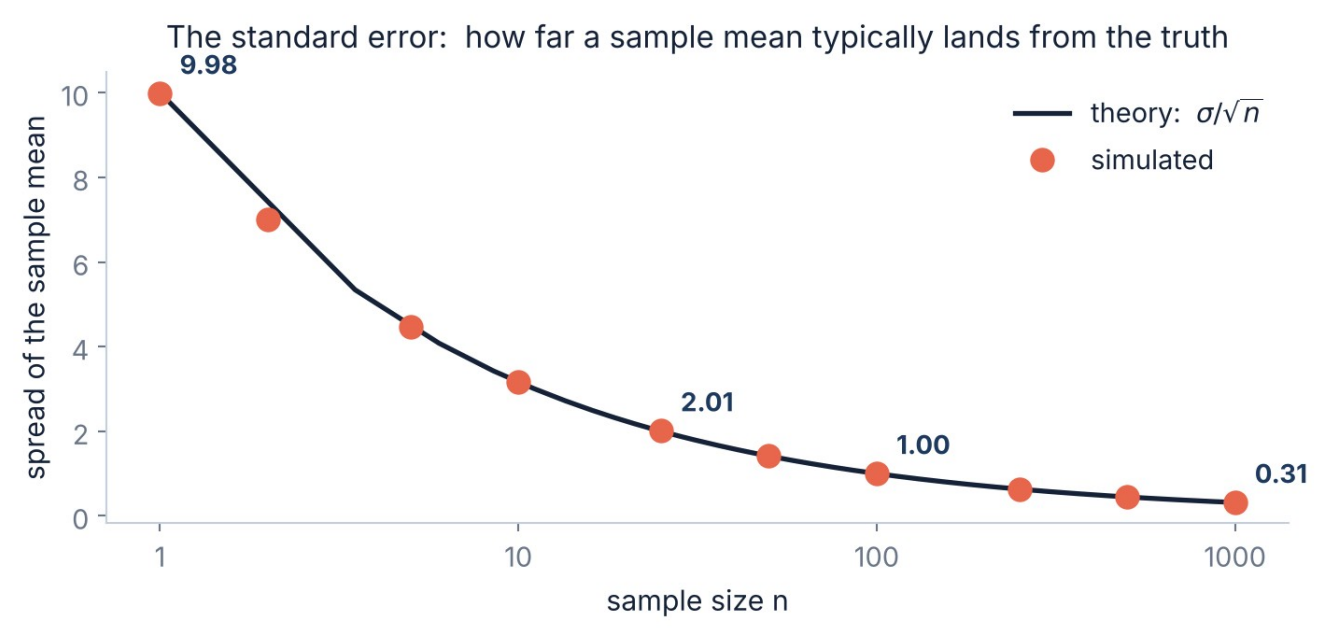
`any population`	No normality assumed. Skewed, bimodal, discrete — it does not matter.
`finite μ and σ`	The one real condition. Section 4 shows what happens without it.
`centred on μ`	The sample mean is unbiased — it is not systematically too high or low.
`σ / √n`	The spread. This is called the STANDARD ERROR, and it is the price list.

“Any population” really does mean any



Look closely at the fourth column. That population takes only three values — 0, 1 and 7 — and its means at n = 30 are visibly ragged and sit slightly off the curve. The theorem is an APPROXIMATION that improves with n, not a switch that flips at some sample size.

The standard error — how wrong your average typically is



Two different numbers

σ is how spread out the DATA is. It does not change when you collect more.

SE = σ/√n is how spread out the MEANS are. It shrinks as you collect more.

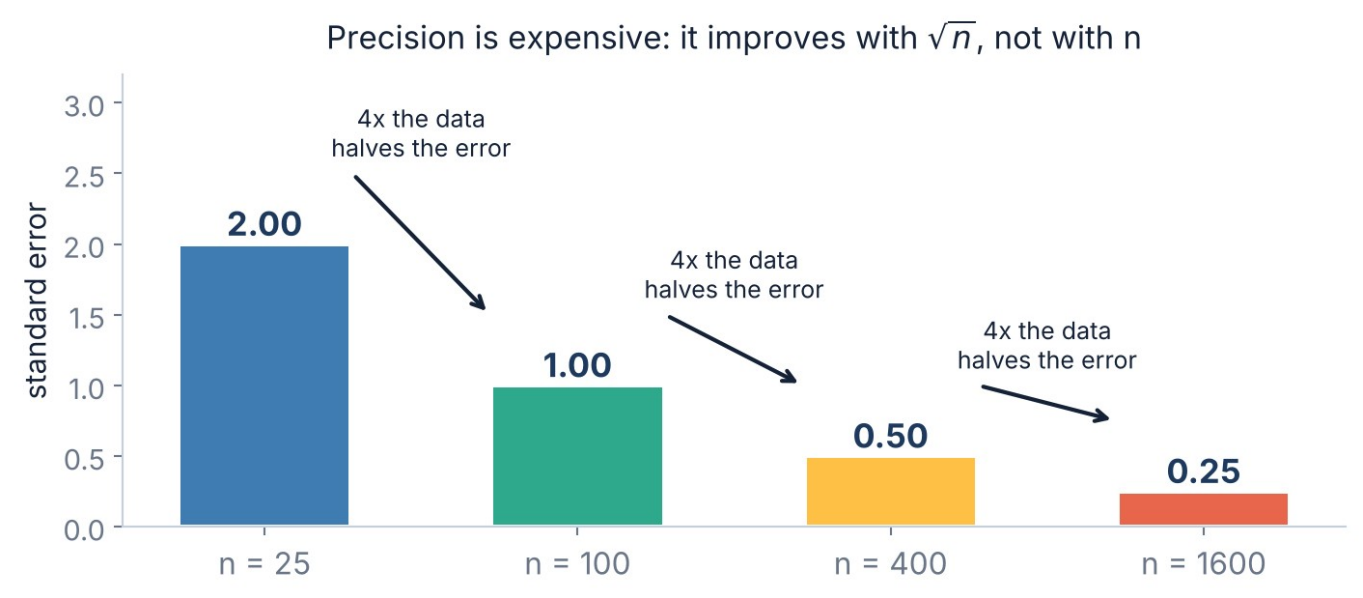
Confusing the two is the second most common error in this topic.

In code

```
sd = data.std(ddof=1)
se = sd / np.sqrt(len(data))
```

ddof=1, because you are estimating σ from a sample — Day 23.

Precision is expensive, and the price is fixed



Four times the data

buys you only twice the precision. Not four times.

To halve your error bars again you need another 4×, and then another. Going from n = 100 to n = 1600 is sixteen times the cost for four times the precision.

Why this is worth knowing

It is the arithmetic behind every “how many users do we need?” conversation. If someone asks you to halve the width of a confidence interval, the answer is four times the data — and that is a budget, not a detail.

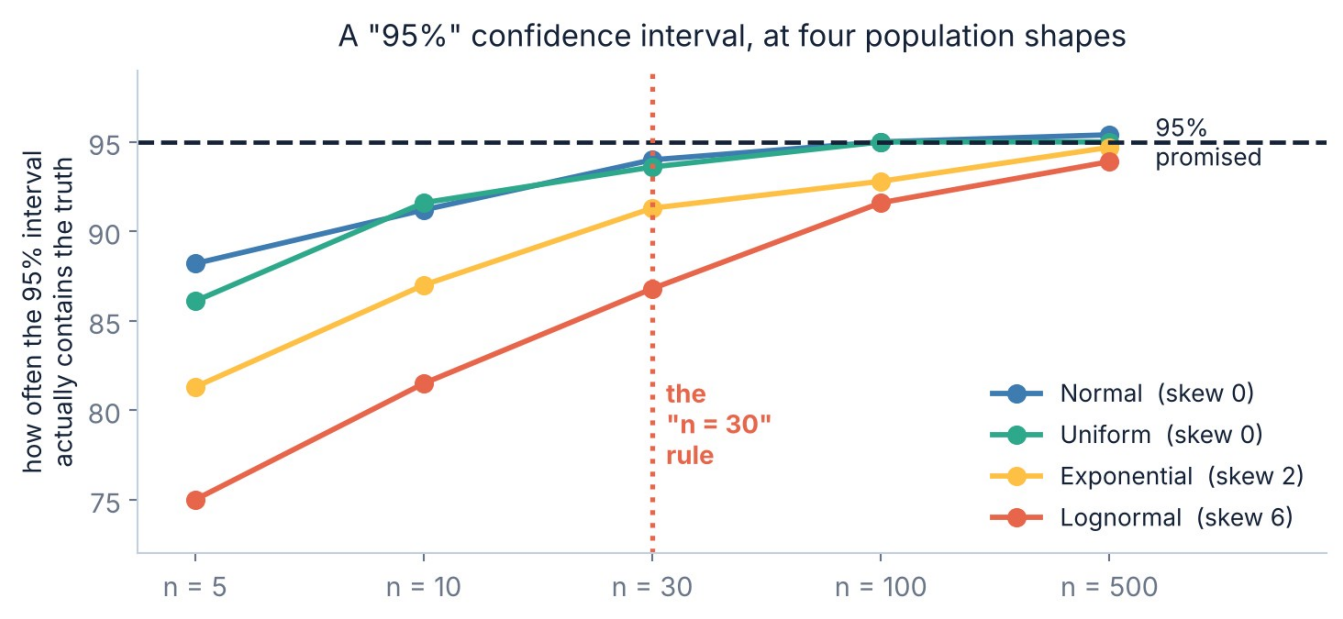
3

How Large Is Large Enough?

The n = 30 rule is not true · 20 minutes

- What the rule claims
- What actually happens
- The number that decides it

⚠️ “n = 30 is enough” is simply not true



What is being measured

Take a sample, build a 95% confidence interval, and check whether it actually contains the true mean. Repeat 4,000 times.

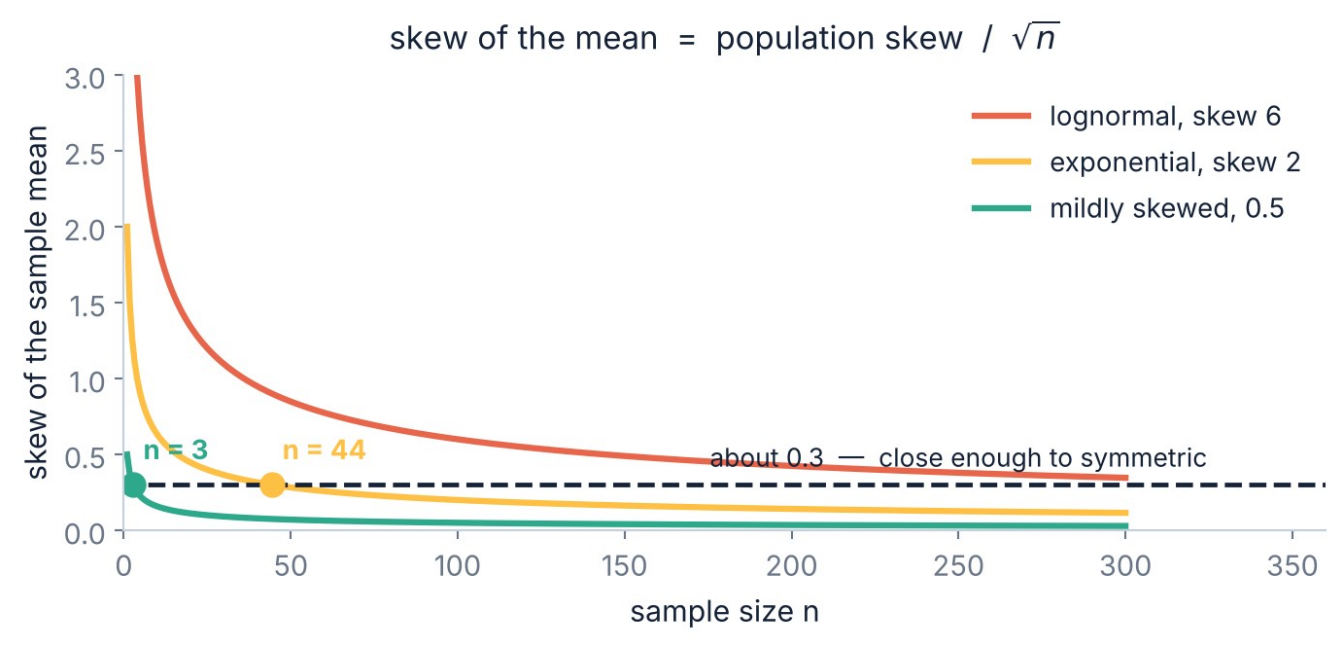
If the method works, 95% of them should.

At n = 30

Population	Covers
Normal	94.0%
Uniform	93.6%
Exponential	91.3%
Lognormal	86.8%

On lognormal data — which is what income, spend, file size and session length all look like — your “95%” interval is really an 87% interval. And even at n = 500 it has only reached 94%.

The number that actually decides how big n must be



skew of the mean = skew / √n

This is exact, not a rule of thumb. The sample mean inherits the population's skew, divided by √n.

So “how large is large enough” has a real answer: large enough that this number is small.

So the honest rule is: measure your data's skew first, then pick n. A mildly skewed population is fine at a handful of observations; a heavily skewed one is still not fine at n = 100. The number 30 appears nowhere in this reasoning.

Working backwards

Aim for a mean-skew near 0.3 and the required n falls out:

- skew 0.5 → n ≈ 3
- skew 2 → n ≈ 44
- skew 6 → n ≈ 400

4

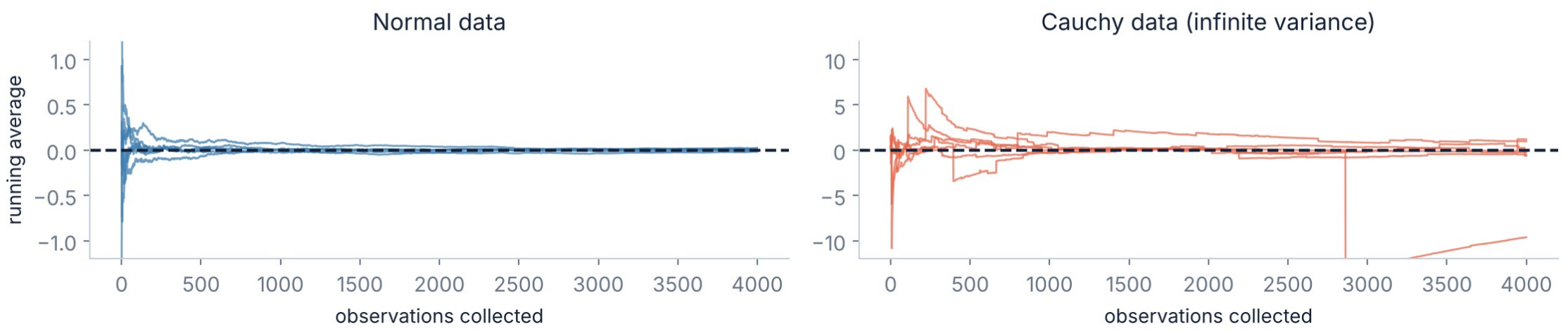
When It Fails

Two ways to lose the guarantee · 15 minutes

- Infinite variance
- Dependence
- The checklist

First failure: when σ does not exist

Six runs each. On the left the average settles. On the right it never does, at any sample size.



The left panel is the law of large numbers

Collect more data and the running average settles down. This is what everyone expects averages to do.

The right panel never settles, at any n

The spread of a Cauchy mean is the SAME at n = 1 and at n = 10,000. Collecting more data buys you nothing at all.

Why that happens, and where you would meet it

The theorem needed a finite σ

A Cauchy has a tail so heavy that its variance is infinite — not large, infinite. σ/√n is then ∞/√n, which is no promise at all.

What happens in practice: one enormous observation arrives and drags the average with it. Then a bigger one arrives. The average never converges because the next outlier is always big enough to move it.

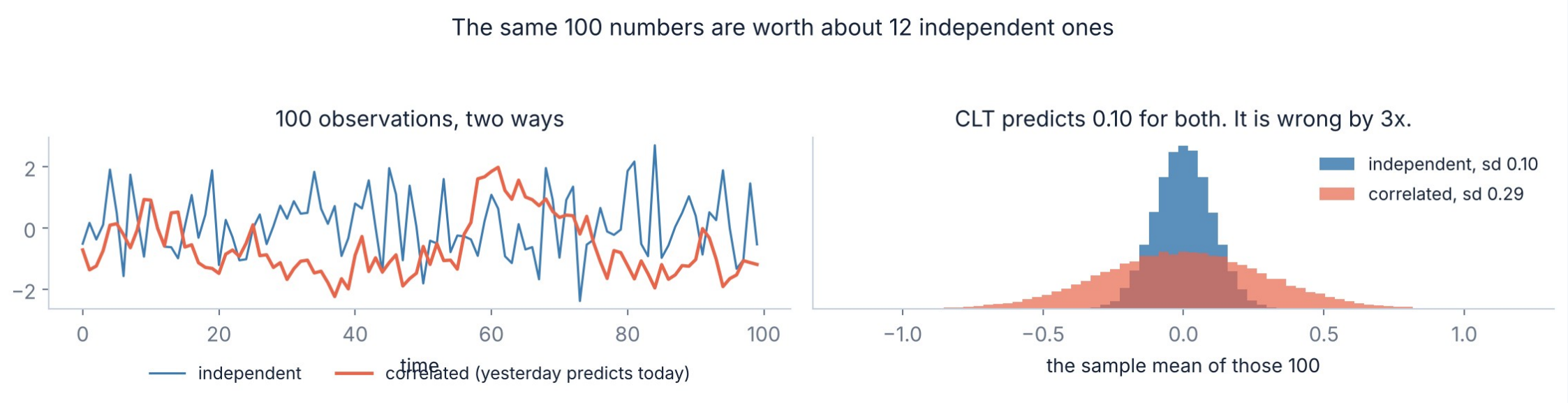
Real data that behaves like this

- Insurance claims — one hurricane dwarfs a decade
- Wealth, and some income distributions
- City and firm sizes
- Financial returns in a crisis
- File sizes and network traffic bursts

The practical test, and what to do instead

Plot the running mean of your own data as you add observations. If it is still jumping after thousands of points, the mean may not be a sensible summary at all — use the median, or a trimmed mean, and say so. Reporting “average claim size” for insurance is a well-known way to be confidently wrong.

Second failure: when the observations are not independent



100 observations, worth about 12

If today's value partly repeats yesterday's, each new reading carries less new information than an independent one would. Here σ/√n says the standard error is 0.10; the truth is 0.29. Every confidence interval built on it is three times too narrow, and every p-value far too small.

Before you lean on the theorem, check four things

1	**Is σ finite?**	Plot the running mean. If it is still jumping after thousands of points, stop.
2	**Are the observations independent?**	Repeated measures on the same people, or anything over time, usually are not.
3	**How skewed is the population?**	stats.skew(data). Then n ≈ (skew / 0.3)² is a defensible starting point.
4	**Am I talking about the data or the mean?**	The theorem only ever licensed a claim about the mean.

Checks 1 and 2 are pass/fail — if they fail, the theorem gives you nothing. Check 3 is a dial, not a gate. Check 4 is free and catches the most common error of all.

5

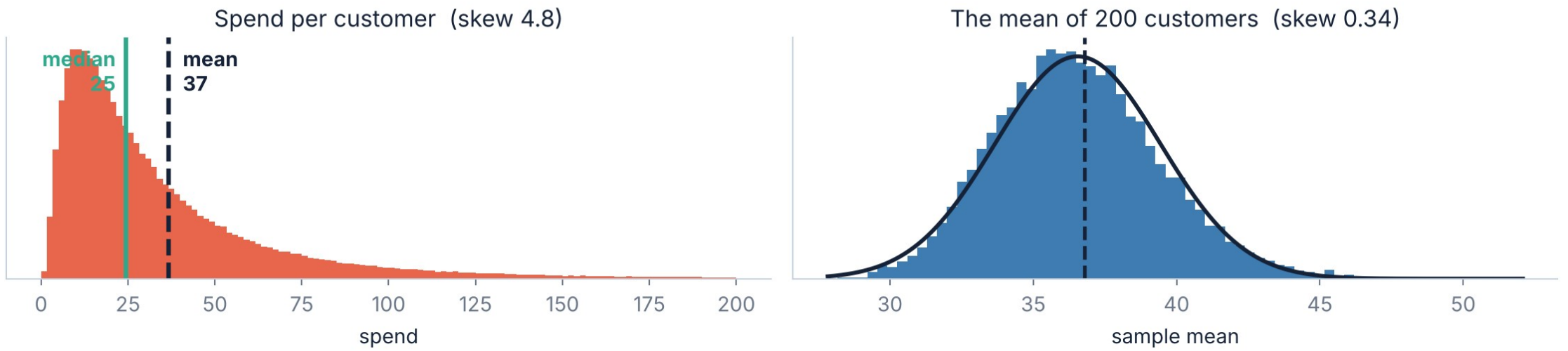
What It Buys You

A worked confidence interval · 20 minutes

- The business question
- The interval
- What 95% actually means

Worked through: what is our average customer worth?

You can never make the left panel normal. You do not need to.



The data will never be normal

Skew 4.75. The typical customer spends 25; the average is 37, pulled up by a long tail. A normality test on this returns p = 4e-18 and always will.

It does not need to be

The mean of 200 has skew 0.34 — 4.75 divided by √200 — and sits almost exactly on the normal curve. That is the object we put an interval around.

Building the interval, one line at a time

```
import numpy as np
from scipy import stats

#  we surveyed 200 customers
n     = 200
xbar  = sample.mean()          # 33.41
sd    = sample.std(ddof=1)     # 34.80

#  --- 1. how wrong is this average likely to be?
se = sd / np.sqrt(n)           # 2.460

#  --- 2. the CLT says the mean is normal, so:
lo = xbar - 1.96 * se          # 28.58
hi = xbar + 1.96 * se          # 38.23

print(f"average spend: {xbar:.2f}  (95% CI {lo:.2f} to {hi:.2f})")

#  the sample itself is wildly non-normal:
stats.skew(sample)             # 3.01
stats.shapiro(sample).pvalue   # 3.6e-18
#  and that does not matter here.
```

ddof=1

estimating σ from a sample — Day 23

÷ √n, not ÷ n

the single most common slip

1.96 is from Day 24

it leaves 2.5% in each tail

The CI is about the MEAN

not about where customers fall

Shapiro fails — and is irrelevant

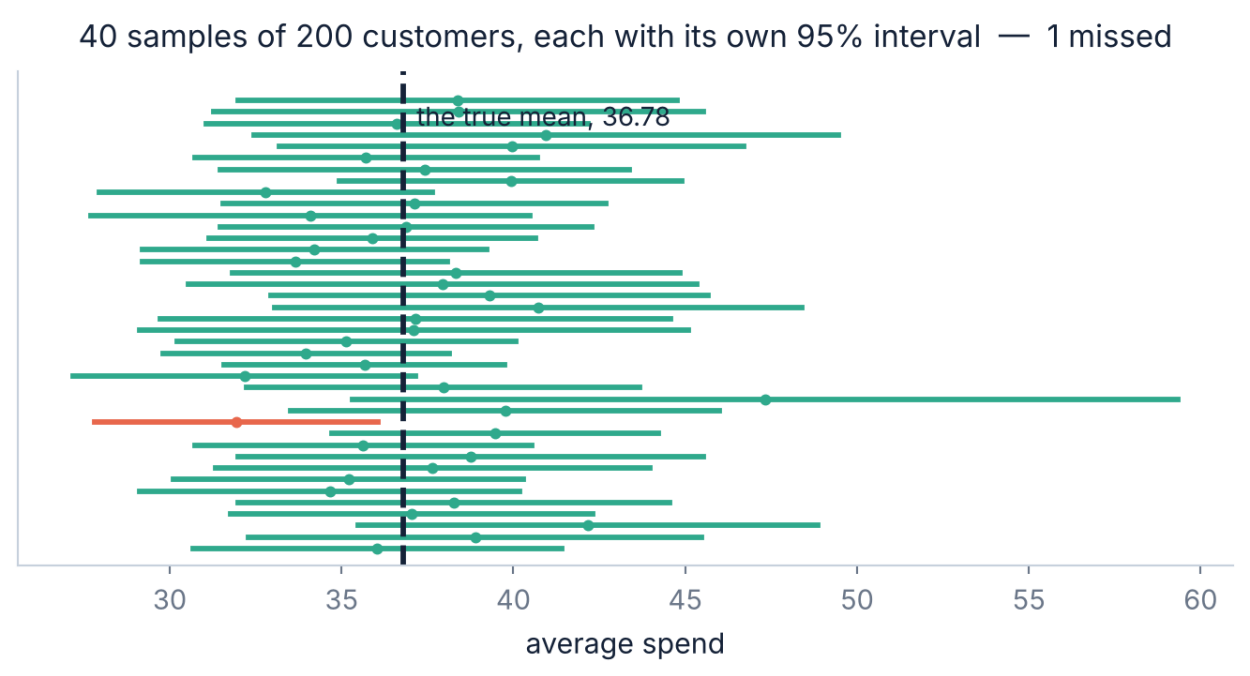
it tested the data, not the mean

True mean is 36.78

the interval caught it, this time

Everything on the left rests on one assumption: that the mean of 200 is normal. The CLT is what lets you assume it, and section 3 is how you check whether 200 was enough.

What “95% confident” actually means



One of these 40 intervals missed. That is the method working as advertised — at 95%, missing about 1 in 20 is the deal you signed. An interval that never missed would mean you were being too cautious.

Read it as a long-run property

Each green bar is one survey's interval. The dashed line is the truth, which in real life you never see.

“95% confident” means: if you repeated the whole survey forever, 95% of the intervals you built would contain it.

⚠️ What it does NOT mean

Not “a 95% chance the true mean is in THIS interval”. Your interval either contains it or it does not — there is no chance left in it.

And not “95% of customers spend between 33 and 41”. That is a statement about people; this is a statement about an average.

Nearly everything ahead of you is standing on this

Confidence intervals

Every ± you have ever seen reported

t-tests and z-tests

“Is this difference real?” — Day 28

A/B testing

Day 25's checkout experiment, done properly

Polling margins of error

±3% is $\sigma/\sqrt{n}$ with $n \approx 1{,}000$

Quality control limits

Day 24's $\pm 3\sigma$ charts, applied to batch means

Regression standard errors

Module 3, every coefficient

Which is why the failure cases are not academic

When the CLT quietly does not hold — heavy tails, correlated observations, a badly skewed population with a small n — none of these methods announce it. They all return a number, and the number is too confident. That is the whole reason section 4 exists.

RECAP

Debug corner — eight CLT mistakes

You did	What goes wrong	Fix
said “n > 30, so my data is normal”	the data never becomes normal	the theorem is about the MEAN
tested the raw data for normality	it fails, and it was never relevant	ask about the mean instead
divided σ by n	the error is far too small	divide by √n
used σ where you needed SE	σ describes people, SE describes the average	SE = σ/√n
trusted n = 30 on skewed data	a “95%” interval covering 87%	$n \approx (\text{skew} / 0.3)^2$
averaged heavy-tailed data	the mean never settles	plot the running mean; use a median
used a time series as if independent	intervals 3x too narrow	effective n, not n
read a CI as “95% of customers”	it is about the mean, not people	say “the average is between…”

Rows 1 and 2 are the same misunderstanding and between them they account for most of the trouble people have with this topic.

Putting it together — a CLT health check

```python
import numpy as np
from scipy import stats

def clt_check(data, n):
    """Can I trust a CLT-based interval from n of these?"""
    d  = np.asarray(data)
    sk = stats.skew(d)

    #  1. does the mean settle at all?
    run = np.cumsum(d) / np.arange(1, len(d) + 1)
    tail = run[len(d)//2:]
    drift = (tail.max() - tail.min()) / abs(d.mean())

    #  2. how skewed will the sample mean be?
    mean_skew = sk / np.sqrt(n)

    #  3. what n would this population actually need?
    n_needed = (sk / 0.3) ** 2

    print(f"population skew   {sk:8.2f}")
    print(f"running-mean drift{drift:8.2%}")
    print(f"skew of mean(n={n}) {mean_skew:6.2f}")
    print(f"n for skew 0.3    {n_needed:8.0f}")
    print("verdict:", "OK" if abs(mean_skew) < 0.3 and drift < .05
          else "use a bootstrap or a bigger n")
```

It checks, it does not assume

three numbers beat one rule

Drift catches heavy tails

a mean that will not settle

skew/√n is the real criterion

not the sample size alone

n_needed is actionable

tells you what to go and collect

It cannot see dependence

no single-sample check can

Run it before every interval

it costs one line

The fifth note matters: nothing you can compute from one sample will tell you whether your observations were independent. Only knowing how the data was collected will.

OVER TO YOU

Practice session — 30 minutes

A **Predict the output** — 8 questions

SE vs σ, √n, skew of the mean

B **Spot & fix the bug** — 4 questions

÷n, the wrong distribution, a bad CI

C **Write the code** — 5 tasks

Simulate, measure, size an experiment

D **Challenge** — 2 tasks

Break the theorem, then rescue it

Your notebook

`Day27_Practice_Questions.ipynb`

- Data, or the mean of the data?
- √n, never n
- Simulate before you trust
- Solutions shared after class

Part D asks you to construct a dataset where the CLT visibly fails, and then to produce a correct interval for it anyway. Both halves are the point.

TODAY IN TWELVE LINES

Recap

1. ⭐ Your DATA and your SAMPLE MEAN are two different distributions.
2. The theorem is about the mean. It says nothing about the data.
3. Sample means are normal, centred on μ, with sd σ/√n — from ANY population.
4. σ/√n is the STANDARD ERROR. σ describes people; SE describes the average.
5. √n law: 4x the data buys 2x the precision. That is a budget.
6. ⚠️ The data never becomes normal. A million salaries are still skewed.
7. ⚠️ “n = 30 is enough” is false. On lognormal data it gives an 87% interval.
8. skew of the mean = population skew / √n. Exact, and it decides n.
9. A defensible rule: $n \approx (\text{skew} / 0.3)^2$.
10. ⚠️ Fails with infinite variance — a Cauchy mean never settles, at any n.
11. ⚠️ Fails with dependence — 100 correlated points were worth 12.
12. Nearly every interval, p-value and A/B test ahead of you rests on this.

Homework

Run clt_check on every numeric column of your Day 21 project and say which ones you could safely build an interval from · Find a published ± figure and work out what n it implies · Take one skewed column and find the n its skew actually demands

Next class — Topic 2.8

Hypothesis testing. Today you learned how far a sample mean typically lands from the truth. Next class turns that into a decision procedure: is this difference real, or is it the √n noise we just measured?